TEXORA · LECTURE SHEET

Fibre Length

Length decides what a cotton can become: the finest spinnable count, the roller settings of every drafting system, and a large part of the price. This sheet covers the length parameters, the distribution behind them, and how each is measured.

TE-3103 · QUALITY EVALUATION OF FIBRE AND YARN · TOPIC 3 · LEVEL 3 TERM 1

1. Why length matters

- **Spinnable count.** Yarn strength comes from fibre-to-fibre friction over the length of overlap. Longer fibres give more overlap per fibre, so a finer yarn still contains enough fibre length in its cross-section to hold together. Extra-long-staple cottons (35 mm and above) spin counts that short cottons cannot reach at any twist.
- **Yarn strength and evenness.** At equal count, longer and more uniform fibre gives stronger, evener yarn with fewer ends down in spinning and weaving.
- **Machine settings.** Roller gauges in drawing, roving and spinning are set just above the effective fibre length; a wrong length figure propagates into drafting waves and irregular yarn.
- **Waste.** The short-fibre fraction is what the comber removes as noil and what floats out of drafting zones to cause thick places. Length *distribution*, not just the mean, controls it.

2. The length parameters — one cotton, several "lengths"

PARAMETER	MEANING
Staple length	The traditional classer's length — the length of a typical, hand-pulled and straightened tuft, judged against standard staples; still the language of trade descriptions (e.g. 1-1/16 in ≈ 27 mm).
Mean length	Average length of all fibres — number-mean or weight-mean according to the method (they differ: weight-mean is higher because long fibres weigh more).
Effective length	The length that characterises the longer, spinning-effective portion of the distribution — historically from the Baer sorter diagram; used to set roller gauges.
Upper-half mean length (UHML)	Mean length of the longer half of the fibres *by weight* — the HVI length figure (Topic 9).

Span lengths	From the Fibrogram: the distance from the clamp at which a stated percentage of fibres still extends — 2.5 % span length approximates staple length; 50 % span length is a mid-distribution point.
Uniformity ratio (UR)	UR % = (50 % span length / 2.5 % span length) × 100. HVI instead reports a **uniformity index (UI)** = (mean length / UHML) × 100 — do not confuse the two.
Short fibre content / index (SFC, SFI)	Percentage (by weight or number) of fibres shorter than 12.7 mm (0.5 in). The single most damaging length statistic for yarn quality.

3. The length distribution and the Fibrogram

A cotton sample contains fibres from under 5 mm (fuzzy seed-coat fibres) to over 40 mm. Plotted as a distribution, the shape — not one number — describes it. The **Fibrogram** is obtained by clamping a randomly caught beard of fibres in the Fibrograph: scanning along the beard, the amount of fibre material falls with distance from the clamp. Percentage span lengths are read from this curve.

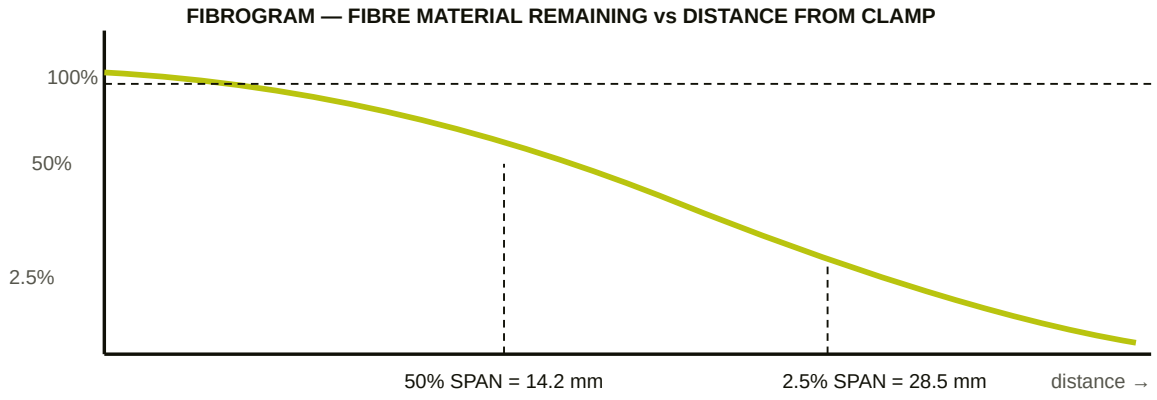


Figure 1 — A Fibrogram. The beard is densest at the clamp and thins as shorter fibres end. Span lengths are the distances at which the stated percentage of fibre still extends.

WORKED EXAMPLE — SPAN LENGTHS AND UNIFORMITY RATIO

A Fibrogram gives 2.5 % span length = 28.5 mm and 50 % span length = 14.2 mm.
UR = 14.2 / 28.5 × 100 = **49.8 %**. Values around 45–50 % are typical for upland cottons; a falling UR at constant 2.5 % span warns that the mid-length population is thinning — expect more drafting trouble even though the "staple length" looks unchanged.

4. Methods of measurement

4.1 Single-fibre measurement

Individual fibres are straightened (without stretching) on a black velvet board over a ruler, or measured with a roller/zwick-type instrument. Hundreds of fibres give the true number distribution — mean, CV, short-fibre percentage by number — but the method is slow and now mainly a reference and research tool. Crimp must be removed by just enough tension to straighten; overstretching wool or textured fibres falsifies length.

4.2 Comb sorter (Baer / Shirley) method

A prepared tuft is transferred through a bed of retractable combs so all fibres start from one baseline; fibres are then withdrawn group by group from the longest downwards and each group weighed. The result is a **weight-based length distribution** — plotted as the sorter diagram — from which mean length, effective length and dispersion are read. Accurate and still the classical reference, but a skilled, slow test (one sample can take an hour).

4.3 Fibrograph (photoelectric beard scanning)

A combed beard specimen is scanned by a light beam; the attenuation along the beard builds the Fibrogram (§3). Fast — the basis of the length measurement inside HVI lines (Topic 9).

4.4 AFIS length measurement

AFIS (Topic 9) individualises fibres and measures length fibre by fibre electro-optically, reporting mean length by number and by weight, and short fibre content by both bases — the fullest routine picture of the distribution.

> **WORKED EXAMPLE — MEAN LENGTH, DISPERSION AND SHORT FIBRE CONTENT**
>
> Individual measurement of 100 fibres gives: 10 mm × 5, 15 mm × 10, 20 mm × 20, 25 mm × 30, 30 mm × 20, 35 mm × 15.
>
> Mean = $\Sigma nL / \Sigma n = (50 + 150 + 400 + 750 + 600 + 525) / 100 = 2475 / 100 =$ **24.75 mm**.
>
> $\Sigma nL^2 / \Sigma n = 658.75$; variance = $658.75 - 24.75^2 = 46.19$; SD = **6.80 mm**; CV = 6.80 / 24.75 × 100 = **27.5 %**.
>
> Short fibre content by number (fibres < 12.7 mm) = 5 / 100 = **5 %**. Note this is a *number* percentage; the same short fibres are a much smaller *weight* percentage — always state the basis.

5. Sampling and test precautions

- Length samples must not lose short fibres: no preliminary hand-cleaning, no vigorous pulling of tufts (the long fibres withdraw first from a sliver or bale — a pulled tuft overstates length).
- Condition the sample (Topic 1); crimp and curl, hence apparent length, change with moisture.

- Report the basis — number or weight — and the instrument. A "mean length" without its basis is meaningless because the weight-mean exceeds the number-mean on the same cotton.
- Fibre breakage in opening and carding shortens the distribution as the material moves through the mill; length measured on raw cotton and on card sliver from the same bale will differ, and both are correct — for different process stages.

Glossary

Staple length — classer's tuft length used in trade. **Effective length** — sorter-diagram length characterising the spinning-effective portion. **UHML** — mean length of the longer half by weight. **Span length** — distance from a Fibrogram clamp at which the stated percentage of fibres extends. **Uniformity ratio** — 50 % span / 2.5 % span × 100. **Short fibre content** — percentage of fibres below 12.7 mm, by weight or number as stated. **Fibrogram** — the curve of fibre material remaining against distance from the clamp in a beard scan.

Key points

• Length governs spinnable count, yarn strength and evenness, roller settings and noil.
• No single number describes length; mean, effective/UHML, span lengths, uniformity and short fibre content each answer a different question.
• Sorter = weight distribution (reference); Fibrograph/HVI = beard scan (fast); AFIS = fibre-by-fibre (fullest).
• Always state the basis (number vs weight) and the process stage sampled.
• Short fibres do the damage: watch SFC even when the staple figure looks good.

Practice

1. Span lengths of a cotton are 2.5 % = 30.0 mm and 50 % = 14.4 mm. Find the uniformity ratio and comment.
2. Groups: 15 mm × 20, 20 mm × 30, 25 mm × 30, 30 mm × 20 fibres. Find the number-mean length.
3. Why is the weight-mean length of a cotton always greater than its number-mean length?
4. A mill sets spinning roller gauges from the raw-cotton staple length, yet card-sliver tests show a shorter effective length. Give two causes and the yarn defect expected.
5. Explain why a hand-pulled tuft from a sliver overstates mean length.

Answer key

1. UR = 14.4/30.0 × 100 = **48.0 %** — within the normal upland range; distribution shape is unremarkable.

2. ΣnL = 300 + 600 + 750 + 600 = 2250; Σn = 100; mean = **22.5 mm**.
3. Each fibre's contribution to the weight average is weighted by its own mass, and mass is proportional to length (at similar fineness). Long fibres therefore count more in the weight mean than in the number mean, pulling it upward.
4. Causes: fibre breakage in opening/carding, and removal of some long fibre into waste / preferential loss — plus the raw figure was a staple (long-end) measure, not an effective length. Gauges set too wide for the actual fibre let fibres float uncontrolled in the drafting zone → drafting wave, thick–thin places and higher yarn irregularity (Topic 13).
5. When a tuft is pulled from a coherent strand, the fibres that withdraw are those gripped longest — the long fibres bridging the pinch point; short fibres stay behind in the strand. The drawn tuft is enriched in long fibre, so its measured mean exceeds the population mean.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib